

SIMATS ENGINEERING

CO2 – ASSESSMENT TOOL 2
Scenario-Based Questions

Course: Deep Learning **Course Code:** MLA04

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1. Unsupervised Training of Neural Networks – Autoencoders

Answer: I would recommend an **autoencoder** because the ECG records are unlabeled. An autoencoder can learn the normal structure of ECG signals without requiring abnormal/normal labels. Abnormal records can then be identified using reconstruction error.

Workflow:

- Collect and preprocess the ECG records, including normalization and noise removal.
- Train an autoencoder mainly on the available ECG patterns.
- The encoder compresses each ECG record into a lower-dimensional latent representation.
- The decoder reconstructs the ECG from the latent representation.
- Calculate reconstruction error between the original and reconstructed ECG.
- Records with unusually high reconstruction error can be flagged for further clinical examination.

2. Autoencoders – Representation Learning

Answer: An autoencoder can reduce the dimensionality of customer purchase data while preserving important patterns. The encoder maps the original high-dimensional purchase features to a smaller latent representation, and the decoder tries to reconstruct the original data.

- Prepare and normalize customer purchase features.
- Train the autoencoder using the original purchase records as both input and reconstruction target.
- Use the encoder output as the compact representation of each customer.
- Apply a clustering method such as K-means to these latent features for customer segmentation.
- The reduced representation can make segmentation more efficient and can remove some redundant information.

3. Width and Depth of Neural Networks – Activation Functions

Answer: Since the facial recognition system performs poorly with a shallow network, the architecture should be made deeper and sufficiently wide to learn more complex facial features. Multiple hidden layers can progressively learn low-level, mid-level, and high-level representations.

- Increase network depth by adding additional hidden layers.
- Use an appropriate number of neurons in each hidden layer so the model has enough capacity.
- Use nonlinear activation functions such as ReLU in hidden layers to learn complex relationships.
- Use a suitable output activation, such as softmax for multi-class face identification.

- Train and validate the deeper model carefully to reduce overfitting, using techniques such as dropout or regularization when appropriate.

4. Restricted Boltzmann Machines (RBMs)

Answer: An RBM can learn hidden relationships between users and movies from viewing or interaction history without requiring manually labeled training data. The visible layer represents user–movie interactions, while the hidden layer learns latent preference patterns.

- Create a user–movie interaction matrix from viewing history.
- Represent the observed interactions in the visible layer of the RBM.
- Train the RBM to learn hidden features representing user preferences and movie characteristics.
- For a user, use the learned hidden representation to estimate preferences for movies not yet watched.
- Rank movies according to predicted preference and recommend the highest-ranked suitable movies.

5. Deep Learning Applications – Manufacturing Defect Detection

Answer: A deep learning system can use representation learning to automatically extract useful features from the sensor readings collected every second and classify products as normal or defective.

- Collect time-stamped sensor readings such as temperature, vibration, pressure, or other relevant measurements.
- Clean, normalize, and organize the readings into suitable time windows.
- Use a neural network suitable for sequential data, such as a recurrent neural network or another deep sequence model.
- The network learns representations of normal and defective operating patterns from training data.
- Use a classification layer to predict the product condition.
- Evaluate the model using measures such as accuracy, precision, recall, F1-score, and a confusion matrix.
- When a high-probability defect is detected, send the result for inspection or corrective action.

Conclusion

The five scenarios demonstrate how unsupervised learning, autoencoders, representation learning, network architecture, activation functions, and RBMs can be applied to real-world prediction, feature-learning, recommendation, and anomaly/defect-detection problems.